A Grammar of Graphics Approach to Managing Numerical Ties in Parallel Coordinate Plots

Denise Bradford

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1 Introduction and Motivation

Parallel coordinate plots (PCPs) are a powerful technique for investigating patterns across multiple attributes (variables) simultaneously (Inselberg 2009). PCPs assign each dimension of an n-dimensional dataset to a vertical axis, arranged in parallel (Wegman 1990). Observations are drawn as polylines connecting single points along each dimensional axes in sequence a PCPs provide perspective on hard-to-visualize multidimensional data that can be used to identify clusters, outliers, and other facets of a dataset.

When multiple observations share the same value in a given dimension, their polylines perfectly overlap, at a changepoint, creating "visual collisions," masking information about the joint distribution between variables as well as obscuring the density along a single axis. The treatment of ties is an aspect not generally addressed in the original parallel coordinate plots of Inselberg (1985) and Wegman (1990).

PCPs were canoncially used to show continuous random variables, but generalizations to for categorical data only exacerbated the problem. Categorical parallel axis plot solutions, such as Sankey[1] and Alluvial diagrams, Categorical PCPs (Pilhöfer and Unwin 2013), and Parallel Sets (Kosara, Bendix, and Hauser 2006) plots all developed (largely independently!) as methods for using parallel axes to represent categorical, bivariate frequency information. Variations on these plots, such as common angle plots (Hofmann and Vendettuoli 2013), address perceptual distortions that arise when line widths are evaluated along the vertical rather than the orthogonal direction, offering an alternative encoding that more faithfully communicates categorical association strengths.

Hammock plots (Schonlau 2003), another type of PCP that accommodates categorical and continuous variables, shows multiple bivariate relationships using parallelograms drawn between axes, obviating the need for tie-breaking and preserving the density information. It is the parallelogram construction that creates the bivariate limitation. Because only bivariate relationships are shown, it is not possible to mentally reconstruct the full joint distribution using a hammock plot.

The development of generalized PCPs (via the software implementation `ggpcp` (VanderPlas et al. 2023)) represents a major step forward in representing the full joint distribution of the data while showing individual observations. One innovation in the `ggpcp` package is the handling of ties in continuous variables: ties are broken and the observations are spread out within the box representing the marginal frequency of each value of the categorical variable. Sorting methods are

[1] Sankey diagrams do not always have parallel axes, but they are visually similar and frequently use parallel axis constructions.

implemented to distribute the tied values in a way that reduces the overall complexity of the plot, as random line-crossings can make PCPs difficult to read and interpret.

This project aims to extend the treatment of categorical variables in generalized PCPs, implementing a solution for the treatment of continuous variables which has the following properties:

- marginal density information (approximately) faithfully represented on the vertical axis
- individual lines can be visually distinguished, at least for data where the number of observations does not induce overplotting.

An immediate consequence of the second objective is that the viewer can, at least in theory, reconstruct the full joint distribution between variables from the plot. We will extend the method used in `ggpcp` to represent categorical data to continuous variables, modifying it to maintain the approximate scaling of the continuous variable, with slight local distortions when tied values occur. In addition, we will develop a consistent visual representation to provide visual cues which indicate the presence of a local distortion due to tied values, while preserving the visual representation of singular values on the parallel axis.

This combination of the advantage of Hammock plots – maintaining the marginal density information – and generalized PCPs, which preserve individual observations and provide the ability to reconstruct the full joint distribution, will enhance GPCPs. Real-world data sets frequently have both categorical and numerical data, and duplicated values in either type of variable. Our extension to GPCPs will facilitate the visual representation of these data sets.

1.1 Related Work

The parallel coordinate plot literature has developed several threads directly relevant to the tie-breaking problem addressed here.

Overplotting and density. Early responses to visual overplotting in PCPs focused on transparency and kernel density estimation rather than positional adjustment. Alpha-blending, as used by Miller and Wegman (1990), allows the density of overlapping lines to become visible through saturation, but sacrifices the ability to trace individual observations. More recent approaches include density-based rendering (Heinrich and Weiskopf 2013) and edge bundling (McDonnell and Mueller 2008), both of which trade individual traceability for aggregate clarity.

Categorical extensions. The categorical parallel axis plot literature developed largely independently of classical PCPs. Parallel Sets (Kosara, Bendix, and Hauser 2006) replaced individual polylines with ribbons whose widths encode bivariate frequencies, enabling unambiguous density reading at the cost of individual tracing. Alluvial diagrams (Brunson 2020) and Sankey diagrams adopt a similar ribbon-based representation. Common angle plots (Hofmann and Vendettuoli 2013) addressed a further perceptual distortion specific to variable-width ribbons: the Müller-Lyer family of line-width illusions causes ribbon widths evaluated along the vertical to appear systematically different from their orthogonal widths. By constraining ribbons to a constant angle, common angle plots recover perceptually accurate frequency judgments. Hammock plots (Schonlau 2003; Schonlau and Yang 2024) occupy a different position: they accommodate both categorical and continuous variables via constant-width parallelograms, but restrict inference to bivariate relationships between adjacent axes.

Perceptual bounds on vertical displacement. A separate thread of the perceptual literature is directly relevant to the question of how much spread the tie-breaking algorithm may safely introduce. The sine illusion, formalized by Day and Stecher (1991) and analyzed in the context of statistical graphics by VanderPlas and Hofmann (2015), establishes that the vertical extent of a line segment is not perceived neutrally: when segments of equal length are arrayed along a curve, those at extrema appear systematically longer than those at intermediate positions. The illusion belongs to the Müller-Lyer family that also underwrites the line-width illusion treated by Hofmann and Vendettuoli (2013) in their development of common angle plots. We draw on this literature in Section 3.3.3 to bound the band-width parameter σ: the same mechanism that distorts perceived line length on a sine wave bounds how much within-band displacement a viewer can sustain before the displacement registers as variance in the underlying data.

Axis and category ordering. A well-studied property of PCPs is that axis order strongly affects both the number of line crossings and the interpretability of patterns (VanderPlas et al. 2023). Minimizing line crossings through reordering is NP-hard in general (Martí and Laguna 2003), but heuristics based on correlation structure and sum of ranking differences provide tractable approximations (Ipkovich, Héberger, and Abonyi 2021). Within a fixed axis ordering, the order of factor levels and the plotting order of individual observations interact with tie-breaking: factor levels ordered to minimize crossings produce fewer extraneous crossings for any given tie-resolution strategy, while the `overplot` parameter in `ggpcp` controls which observations are rendered on top when lines overlap (VanderPlas et al. 2023). These considerations interact directly with the hierarchical sorting approach developed in this proposal and are discussed further in Section 2.2.

Computational quality metrics. Dennig et al. (2021) formalized a suite of screen-space quality metrics for Parallel Sets—including ribbon overlap, crossing angle, orthogonality, and ribbon width variance—collectively termed ParSetgnostics. These metrics provide an empirical basis for comparing and optimizing parallel axis layouts without requiring user studies, and form the basis for the computational benchmarking component of our evaluation plan (Section 4.3).

1.2 Categorical Variables and Ties in ggpcp

The `ggpcp` package currently addresses categorical ties through a combination of sorting and tie-breaking algorithms. The package implements hierarchical sorting through the `pcp_arrange(data, method, space)` function, which orders observations based on a hierarchical application of variable values. The `method` parameter determines the sequence in which variables are considered when resolving ties in the arrangement. The `space` parameter specifies the proportion of the y-axis dedicated to empty space between levels of categorical variables (Figure 1).

```r
# 1. Modular Data Pipeline (The correct Step 1, 2, 3)
# Note: 'pcp_data' is used here as a variable name for the transformed object.
pcp_data <- mtcars %>%
  mutate(across(c(cyl, am, gear, carb), as.factor)) %>%
  pcp_select(cyl, am, gear, carb) %>%     # Step 1: Reshape data
  pcp_scale(method = "uniminmax") %>%     # Step 2: Scale axes
  pcp_arrange(method="from-left")         # Step 3: Break Ties,
                                          #         sorting from left
```

The ability to follow individual observations is key to being able to reconstruct the full joint distribution across parallel axes. `ggpcp` creates equally spaced points within each category that span the portion of the vertical axis dedicated to the category (preserving the marginal frequency information, which is optionally emphasized with `geom_pcp_boxes`). The top left plot in Figure 1 shows the PCP without this categorical tie-breaking approach. However, `ggpcp` goes one step further

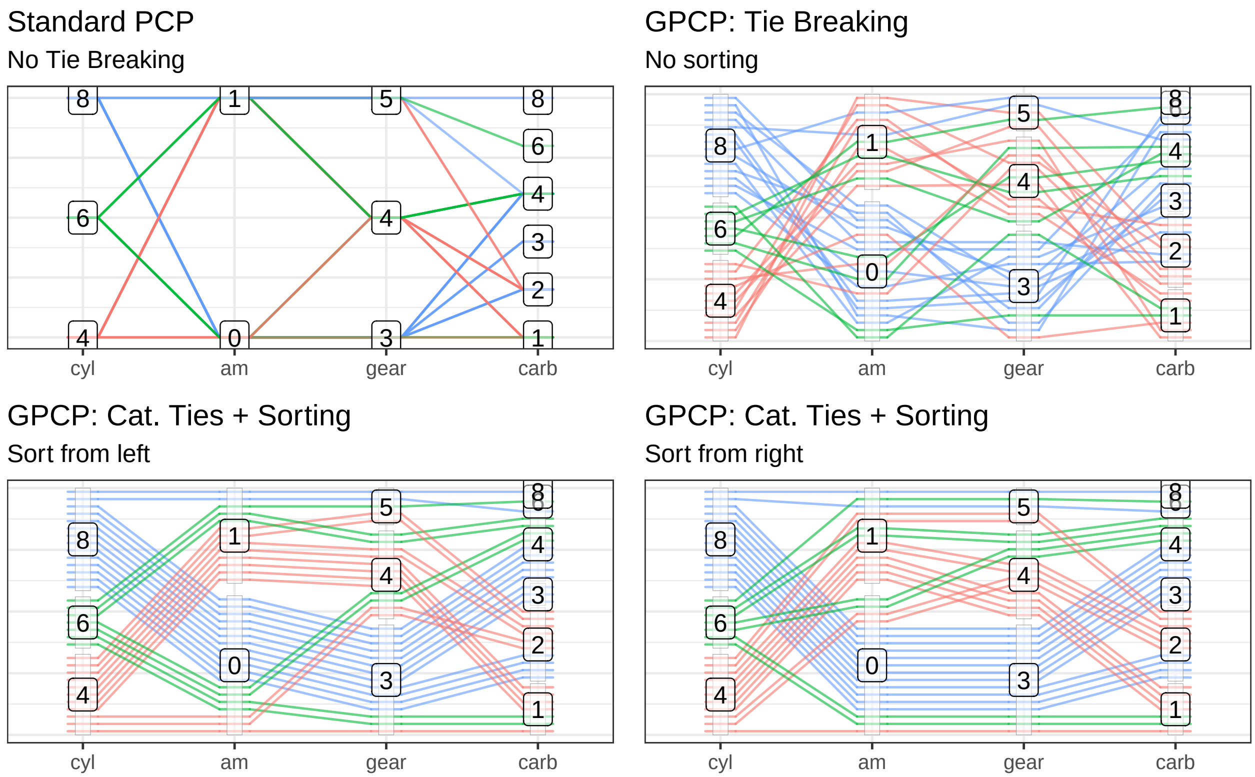

Figure 1: Comparison of tie-breaking methods for categorical variables in parallel coordinate plots using mtcars data. Standard PCP (top-left) shows overlapping lines without tie-breaking. GPCP with tie-breaking but no sorting (top-right) spreads observations evenly within categories. GPCP with sorting from left (bottom-left) and from right (bottom-right) apply hierarchical sorting to minimize line crossings, with light gray boxes indicating category groupings.

by using hierarchical sorting (from the left, right, or both) to minimize unnecessary line crossings induced by the treatment of tied categorical variables. The absence of the hierarchical sorting approach is emphasized in the top right plot in Figure 1, while the bottom left and right plots show sorting from the left and right respectively. This hierarchical sorting approach serves as a form of "external cognition," reducing the cognitive load required to "untangle" and group crossed lines. The sorting also ensures that only necessary crossings[2] occur, minimizing visual clutter.

It is worthwhile to examine how a viewer would follow a line across the plot in detail, using a generalized PCP with both numeric and categorical variables to illustrate the challenges introduced when numerical ties are present.

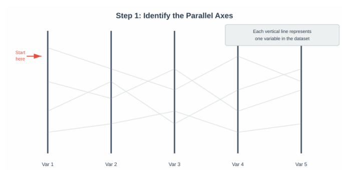

Figure 2: The parallel axes form the structural framework of the visualization.

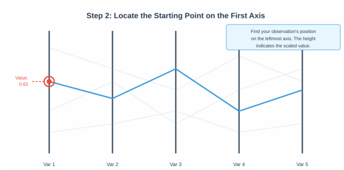

Figure 3: The starting point is identified on the first axis with its corresponding value.

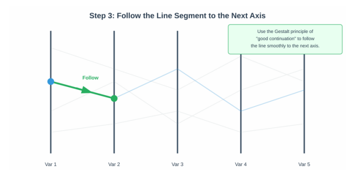

Figure 4: Following the line segment uses the Gestalt principle of good continuation.

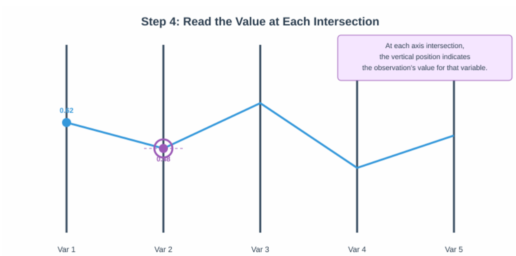

Figure 5: Values are read as approximate scaled value relative to the total range.

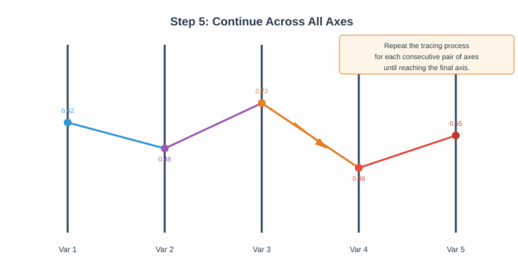

Figure 6: The complete traced path reveals the observation's values across all variables.

- Step 1: Identify the Parallel Axes (Figure 2).
  Begin by identifying each vertical axis in the plot. Each axis represents one variable from the dataset. The axes are typically arranged from left to right, and the order may be determined by the data analyst to highlight specific relationships or minimize visual clutter.
- Step 2: Locate the Starting Point (Figure 3).
  Find the observation of interest on the leftmost axis. The vertical position indicates the scaled value of that observation for the first variable. If you are examining a highlighted or color-coded observation, look for its distinctive marker at this starting position.
- Step 3: Follow the Line Segment (Figure 4).
  Trace the line segment from the starting point to its intersection with the next axis. The human visual system naturally follows smooth, continuous paths due to the Gestalt principle of good continuation, which allows viewers to perceive connected line segments as a single, piecewise continuous line, making it easier to track observations across multiple parallel axes.

[2] A crossing is necessary if it is induced by the order of x-axis variables and scaled values, rather than by the treatment of categorical variables.

- Step 4: Read Values at Intersections (Figure 5).
  At each axis intersection, the vertical position of the line indicates the observation's value for that variable. Read these values to understand how the observation changes (typically, relative to the other observations) across different dimensions of the data. The slope of line segments between axes provides information about the relationship between consecutive variables for that specific observation.

- Step 5: Continue Across All Axes (Figure 6).
  Repeat the tracing process for each consecutive pair of axes until reaching the rightmost axis. By following the complete path, you obtain a comprehensive view of how that particular observation behaves across all measured variables. The vertical positions along each axis represent scaled values that can be interpreted as quantiles when the data are appropriately transformed. This enables identification of unique characteristics, cluster membership, or outlier status.

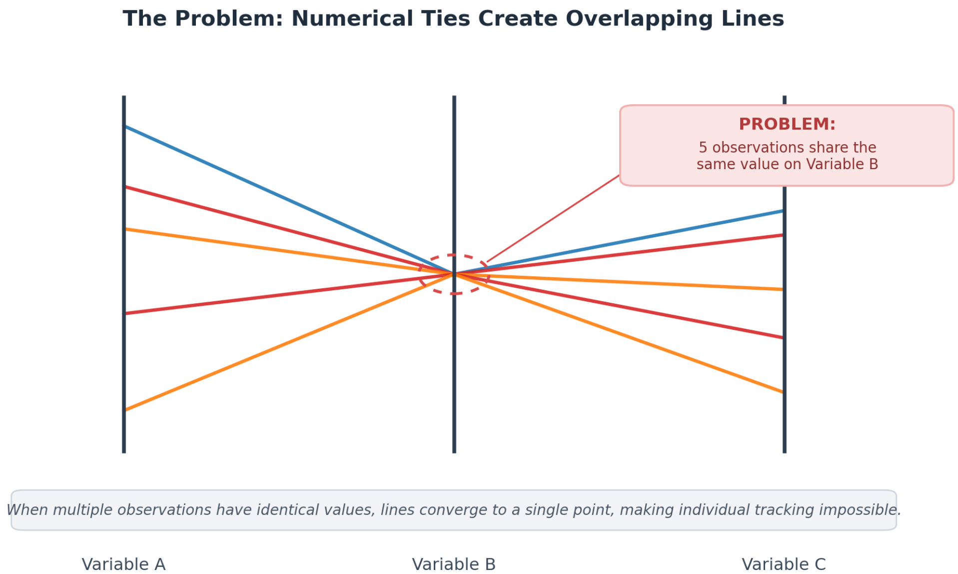

Figure 7: Numerical ties cause lines to converge at a single point. Line identity at the tie can only be recovered when lines arrive with sufficiently distinct colors; here the blue line is distinguishable by opacity, but the cyan and red lines are not cleanly separable, and the values each observation takes on Variables B and C are not uniquely determined from the plot alone. A revised version of this figure will use 2–3 categorically distinct colors to make this failure mode explicit.

- Step 6: Ties in Numbers — A Fork in the Road (Figure 7).

What was once a single point now becomes the starting point for two separate line segments that go in different directions. The Gestalt principle of good continuation doesn't work anymore when two or more lines come from the same point on an axis. In this configuration, color is the only channel that can provide separate line identity across the tie — and even then, only when the palette is carefully chosen. The visual continuity that made tracing easy in earlier steps breaks down exactly where it is needed most. The observer cannot tell if the original observation follows the upper branch or the lower branch of the fork unless there are more visual cues, like color coding, clear labels, or a systematic tie-breaking arrangement. This lack of clarity is not just an aesthetic problem; it also makes it harder to do the main analytical job of the parallel coordinate plot, which is to show patterns that would be hard to see in tabular data by following individual observations through high-dimensional space.

1.3 Addressing Numerical Ties

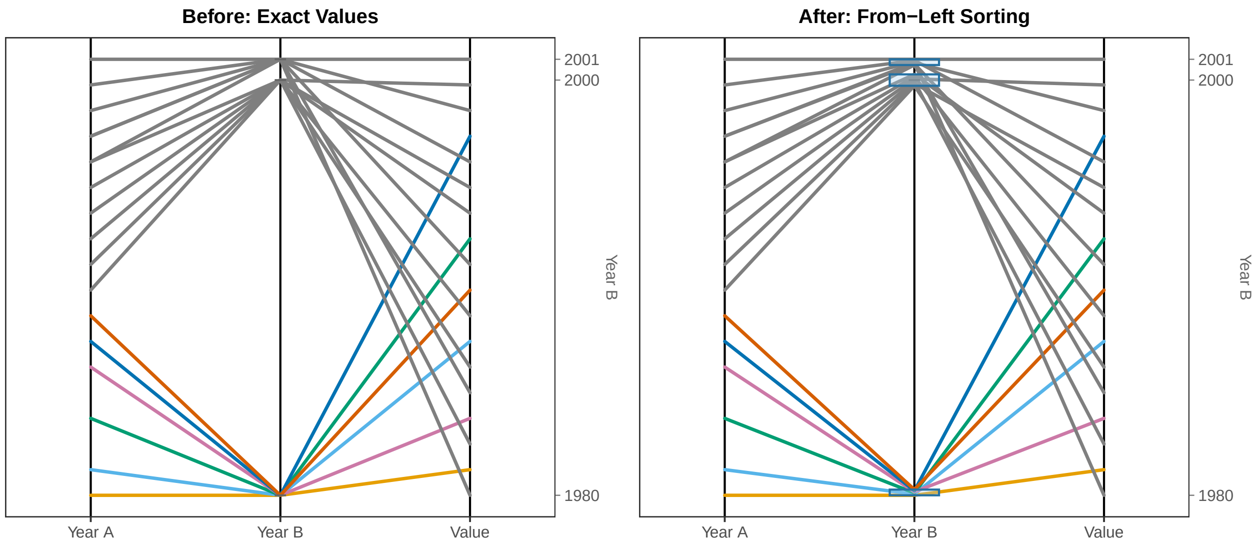

Figure 8: Comparison of a standard PCP with overlapping ties (left) and the proposed tie-breaking algorithm with `from-left` hierarchical sorting (right). Tied observations on Year B are spread within a resolution-based interval (blue boxes) and ordered by their Year A values, eliminating within-band crossings. A visible gap is preserved between the tie bands for 2000 and 2001: separation between bands is essential for the Gestalt grouping argument to hold, since adjacent bands that touch would visually merge and the reader would lose the ability to distinguish distinct values. The band width is therefore set strictly below the global axis resolution δ_j. Remaining crossings are necessary crossings induced by the data structure rather than artifacts of tie-breaking.

Two implementation details in Figure 8 deserve explicit comment because both are load-bearing for the Gestalt argument that motivates the spreading. First, each tie band must be strictly narrower than the gap to its nearest neighbor: if the band around $v = 2000$ extended up to meet the band around $v = 2001$, the two values would visually fuse, defeating the separation the spread was meant to make visible. The band width in the figure is therefore set to a fraction of the axis resolution δ_j, leaving a deliberate gap between adjacent bands. Second, the reader must be able to see that gap — not merely that it exists in the coordinate system. This is a Gestalt-of-proximity requirement: if ties at adjacent values are rendered without visible separation, the grouping cue that "observations within a band share a value" competes with the unintended cue that "bands adjacent to one another belong to the same group." The mathematical formalization of this separation requirement appears in Step 3 of the algorithm (Section 3.3.3).

The `ggpcp` package implements a sophisticated tie-breaking algorithm for categorical variables that maintains individual observation traceability. The approach spaces observations evenly within categorical levels:

> "All observations are spaced out evenly. This results in a natural visualization of the marginal frequencies along each axis (additionally enhanced by the light gray boxes

grouping observations in the same category) that is not as prominent in the previous three panels. The ordering of the observations within the level is such that a minimal number of line crossings occurs between the axes.” (p. 11)

Even spacing alone, however, is not sufficient: without a principled ordering within each tie band, lines departing toward different positions on the neighboring axis will cross inside the band, reproducing the very confusion the spreading was meant to resolve. The hierarchical sorting approach that resolves this is described in detail in Section 2.2; the spacing formula used for categorical levels is:

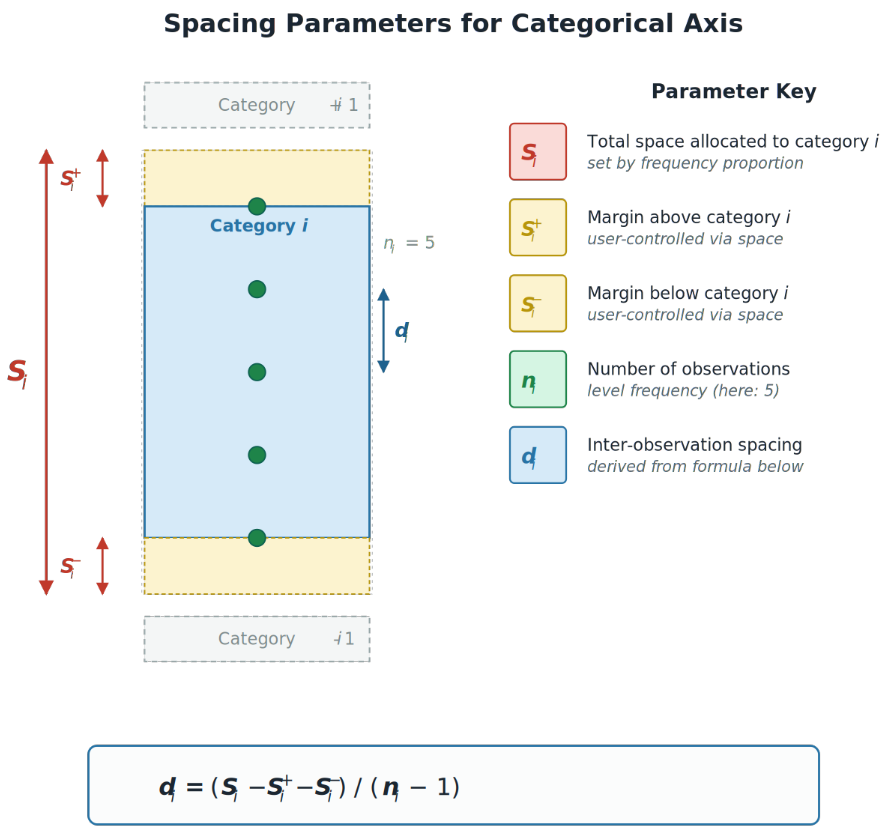

Figure 9: Spacing parameters for a categorical axis. Each level i is allocated total height S_i, with margins S_i^+ above and S_i^- below reserved for inter-level separation. The remaining vertical extent is divided into $n_i - 1$ intervals to seat n_i observations at uniform spacing d_i. The parameter-key panel on the right indicates how each quantity is determined: S_i is set by the level’s frequency proportion under a user-chosen total budget, $S_i^{\pm}$ are user-controlled via the `space` argument, n_i is the level frequency, and d_i is derived from the formula at the bottom. The numerical analog of this figure (Figure 10) shares the same layout, color encoding, and formula structure; the differences are confined to how each parameter is set.

$$d_i = \frac{S_i - S_i^- - S_i^+}{n_i - 1}$$

where:

- S_i is the total space allocated to category i
- S_i^- is the spacing below category i
- S_i^+ is the spacing above category i
- n_i is the number of observations in category i
- d_i is the optimal spacing distance between consecutive observations

The spacing formula above is the categorical workhorse: each level i is allocated total height S_i, lower and upper margins S_i^- and S_i^+ are subtracted to enforce inter-level separation, and the remaining vertical extent is divided into $n_i - 1$ intervals to seat n_i observations. The proposed numerical framework adopts the same five quantities — total allocated space, separation above and below, count, and inter-observation spacing — and the same formula structure, but redefines what determines each parameter. Where the categorical algorithm assigns S_i in proportion to the level frequency under a user-chosen total budget, the numerical algorithm anchors $S(v) = \delta_j$ to the global axis resolution, so the per-value allocation is the same for every v. The categorical S_i^- and S_i^+ are user-controlled through the `space` argument; their numerical analogs are derived from the separation parameter σ and fixed at $(1-\sigma)\cdot\delta_j/2$. The count n_i becomes the count of observations tied at value v, and the spacing formula

$$\Delta(v) = \frac{S(v) - S^+(v) - S^-(v)}{n(v) - 1}$$

is structurally identical to its categorical counterpart $d_i = (S_i - S_i^+ - S_i^-)/(n_i - 1)$.

Figure 10 displays these quantities using the same parameter-key conventions as Figure 9. The intent is for the reader to be able to lay the two figures side by side and read the algorithm as a single procedure with type-specific instantiations of the same five parameters, rather than as two separate algorithms that happen to share an idea. The differences — where S_i is set by frequency and $S(v)$ is set by resolution; where $S_i^\pm$ is user-controlled and $S^\pm(v)$ is derived from σ — are the substance of the innovation, and they are visible in the figures only if the notation is shared.

A key visual difference emerges when connecting categorical to numerical variables. Schonlau and Yang (2024) observe:

> "When many observations have the same value for a categorical and an adjacent numerical variable, the corresponding area looks like a triangle… Notice the lines/boxes between the variables hospitalizations and comorbidities in the GPCP (Figure 13) and hammock plots (Figure 2). Most of the observations are in the boxes leading from hospitalizations=0 to either comorbidities=0 or comorbidities=1. This is far more obvious in the hammock plot than in the GPCP plot." (p. 19)

Whether the transition takes a triangular or rectangular form is perhaps less important than what it communicates about the data itself. The more interesting question is whether the visualization preserves density information while still allowing viewers to follow individual observations. When observations flow from a categorical box toward numerical values, tie-breaking on the numerical axis shapes how clearly we can perceive the underlying distribution.

The `ggpcp` package tackles this challenge through its tie resolution parameters. The `space` parameter controls vertical spacing between tied values, and the `method` parameter governs how those values are ordered. Used well, these parameters spread observations along the numerical axis much like a rug plot would, giving readers a sense of density while keeping individual paths traceable. This represents a different set of trade-offs than hammock plots, which use constant-width parallelograms to show marginal densities directly. Rather than foregrounding aggregate density, `ggpcp` keeps individual observation lines intact and relies on thoughtful spacing to convey distributional structure. The result is a visualization that supports both seeing the forest and examining particular trees.

The preceding review of categorical tie-breaking, together with the related work in Section 1.1, sets the stage for a formal specification of the research questions this proposal addresses.

1.4 Research Questions

The preceding review motivates three formal research questions that structure the remainder of this proposal.

RQ1 (Algorithm correctness). Does the `tie_spread` algorithm distribute tied observations within resolution-based intervals in a way that (a) preserves marginal frequency information on the vertical axis and (b) eliminates within-band line crossings through hierarchical sorting?

RQ2 (Frequency perception). Do viewers make more accurate frequency judgments—for magnitude estimation, ordinal comparison, and ratio tasks—when examining a GPCP with numerical tie-breaking than when examining an equivalent hammock plot?

RQ3 (Individual traceability). Does numerical tie-breaking in GPCPs improve viewers' ability to follow a single observation across all axes, relative to a standard PCP without tie-breaking, without sacrificing the density information conveyed by the marginal axis representation?

RQ1 is addressed computationally through the algorithm development and visual inspection phases (Phase 1 and 3, Section 4.2). RQ2 and RQ3 are addressed through the perceptual validation study (Phase 2). We pre-specify directional hypotheses for RQ2 and RQ3: we anticipate that GPCP with tie-breaking will produce lower absolute percentage error on magnitude estimation tasks (RQ2) and higher tracing accuracy (RQ3) than the respective baselines, with effect sizes consistent with a medium Cohen's d based on prior graphical perception literature.

2 An Approach to Numerical Ties

Currently, numerical variables disrupt some of the best innovations in the GPCP approach by making it impossible to track an observation across the plot and by obscuring marginal density information through overplotting. The remaining two chapters of the dissertation will address the handling of ties on numerical axes and assess the utility of visual cues that can be paired with the tie breaking method to indicate that values are approximately accurate spatially and equivalent numerically.

The `ggpcp` package does not currently provide a mechanism for resolving these tied values. We propose `tie_spread`, an algorithm that distributes observations with identical values along the vertical axis, transforming overlapping lines into a visually resolvable spread. Consistent with tidyverse conventions and the tidyselect grammar, the implementation allows users to specify tie-breaking behavior selectively across axes—applying spreading to some variables while preserving exact positioning on others. Sensible defaults balance visual clarity against faithful representation of the underlying data, ensuring that users can immediately benefit from improved legibility without extensive parameter tuning. The following section details the design and implementation of this approach.

2.1 The Problem: Overlapping Lines

When multiple observations share the same value on a numerical axis, their polylines converge to a single point on that axis, producing perfectly overlapping line segments on either side. This convergence creates two compounding problems: individual observation traceability is lost, because the viewer cannot distinguish which incoming line corresponds to which outgoing path, and marginal density information is obscured, because a cluster of ten tied observations is visually indistinguishable from a single one. The problem is particularly acute for integer-valued or coarsely-measured variables, where ties are structurally common rather than incidental (Figure 7).

The `tie_spread` algorithm resolves tied observations by distributing them evenly within a vertical allocation anchored to the axis resolution δ_j — the smallest distance between any two distinct values on the axis. Each unique value v on the axis receives an allocation of total height

$$S(v) = \delta_j,$$

with margins $S^+(v) = S^-(v) = (1-\sigma) \cdot \delta_j/2$ reserved at the top and bottom of the allocation to preserve a perceptible gap between adjacent value bands. The remaining interior of height $\sigma \cdot \delta_j$ is the dot-placement region: tied observations at value v are spread evenly from the lower to the upper boundary of this interior at uniform spacing $\Delta(v) = (S(v) - S^+(v) - S^-(v))/(n(v)-1)$. This directly mirrors the categorical tie-breaking approach already implemented in `ggpcp`, extending it to numerical variables with the same five quantities (S, S^+, S^-, n, and inter-observation spacing) and the same formula structure (Figure 10). The two frameworks differ only in what determines each parameter: where the categorical algorithm allocates space proportional to level frequency, the numerical algorithm allocates the same δ_j to every value. Marginal density on a numerical axis is therefore encoded by the *dot density within each band* rather than by the *vertical extent of each band* — a design choice that preserves the natural numerical ordering on the axis and avoids the perceptual distortion that would arise from variable-height bands at fixed numerical positions.

Figure 11 shows the parameter spec of Figure 10 applied to a small parallel coordinate plot. Variable B carries three tied groups at $v \in \{0.8, 0.5, 0.2\}$, with $n(v) = 5, 3, 2$ observations respectively. All three bands have the same vertical allocation $S(v) = \delta_j$, regardless of n — the algorithm encodes marginal density through dot density within the band, not through band height. The figure illustrates `from-right` processing — the complement of the `from-left` mode used in Figure 8 — so that observations within each band are placed in the order determined by their values on Variable C, the right neighbor. The polylines emerging from the band onto Variable C therefore carry no avoidable crossings; the within-band rank is the visible signature of the hierarchical sorting introduced in Section 2.2. Together the two figures illustrate that the algorithm is symmetric: the same construction applies under either processing direction, with the role of the reference axis swapped.

2.2 Hierarchical Sorting for Minimal Crossings

Spreading tied values evenly across a resolution-based interval is a necessary first step, but it is not, by itself, sufficient to eliminate visual knots. When the order of assignment within the tie band is arbitrary, lines departing toward different positions on the neighboring axis will cross inside the band, reproducing the very confusion the spreading was meant to resolve. The `ggpcp` package resolves this through hierarchical sorting that mirrors the approach already used for categorical variables.

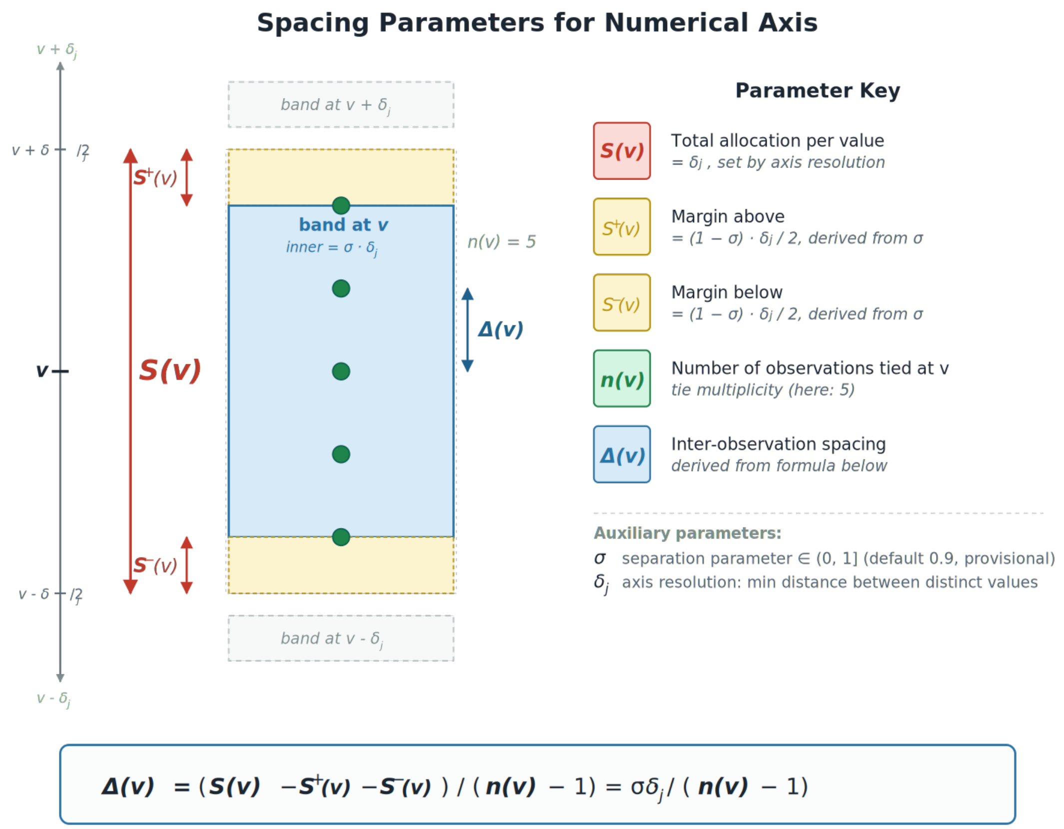

Figure 10: Spacing parameters for a numerical axis, displayed in the same layout as Figure 9 so the two frameworks read as a single procedure with type-specific instantiations. Each unique value v receives a vertical allocation of height $S(v) = \delta_j$, with margins $S^+(v)$ above and $S^-(v)$ below — both fixed at $(1 - \sigma) \cdot \delta_j/2$ — that preserve the inter-band gap. The remaining inner region of height $\sigma \cdot \delta_j$ is divided into $n(v) - 1$ intervals to seat $n(v)$ tied observations at uniform spacing $\Delta(v)$. The mapping to Figure 9 is direct: $S_i \leftrightarrow S(v)$, $S_i^{\pm} \leftrightarrow S^{\pm}(v)$, $n_i \leftrightarrow n(v)$, $d_i \leftrightarrow \Delta(v)$, and the spacing formula at the bottom is structurally identical. The innovation, visible in the parameter-key annotations, is in how each quantity is determined: $S(v)$ is anchored to the global axis resolution rather than to frequency, and $S^{\pm}(v)$ are derived from the separation parameter σ rather than user-controlled.

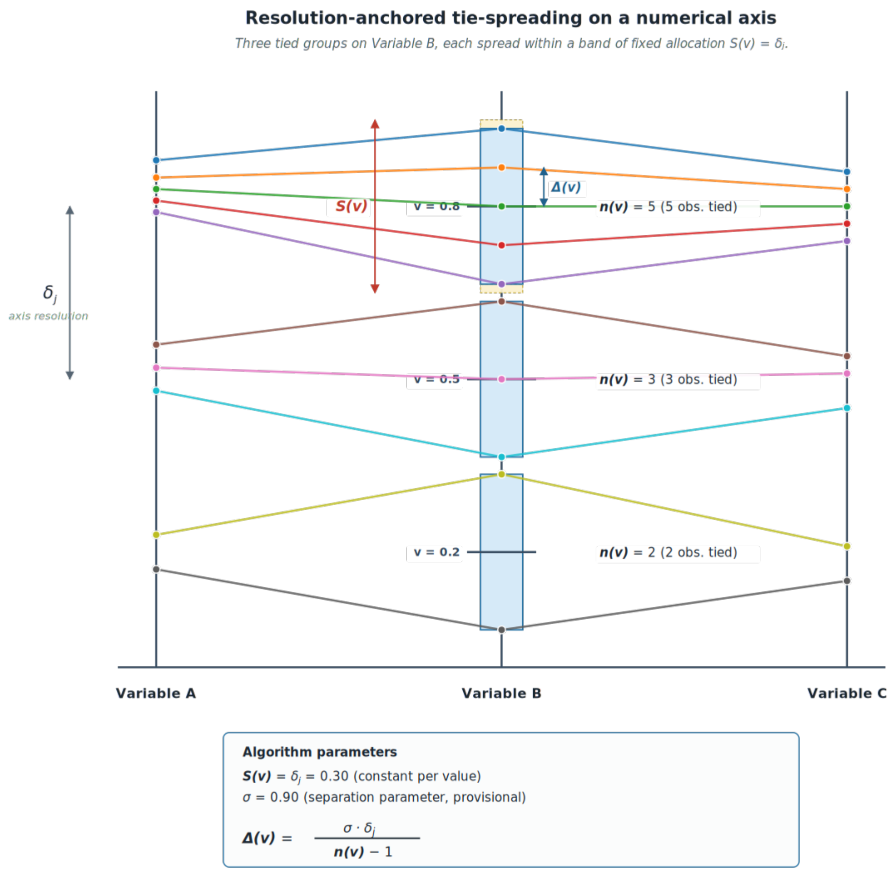

Figure 11: The algorithm of Figure 10 applied to a small parallel coordinate plot. Three tied groups on Variable B, with $n(v) = 5, 3, 2$, are spread within bands of identical vertical allocation $S(v) = \delta_j = 0.30$. The top band carries the parameter-reference annotations ($S(v)$, $\Delta(v)$, the yellow $S^{\pm}(v)$ margin strips); the mid and bot bands are shown as plain inner regions to keep the visual uncluttered. The solid horizontal line and "v = ·" label at each band's center mark the actual tied value, against which the within-band spread is the algorithm's displacement of each tied observation. Within each band, the from-right ordering rule places the observation with the highest value on Variable C at the top of the band and the lowest at the bottom, so that polylines emerging from the band onto Variable C run in monotone order. Polylines are drawn as straight segments between axes; distinct colors per observation allow individual polylines to be traced through the tied region — the legibility property the algorithm is designed to preserve.

For the `from-right` processing mode — the mode directly applicable to Figure 8 — each tied observation $i \in T_j(v)$ is ranked by its adjusted position on the right neighbor axis, $\kappa_i = \tilde{y}_{i,A_j}$ (see Equation 9). Positions within the tie band are then assigned in the same rank order, from lowest to highest. This monotone rank-matching guarantees that no two lines cross within the tie band: the observation that arrives lowest on the right axis is placed lowest in the band, the next-lowest observation is placed next, and so on. Any crossings that remain after sorting are *necessary* crossings — those induced by the data structure itself rather than by the mechanics of tie-breaking.

By ordering observations within a tie group based on their values on adjacent axes, the package leverages the Gestalt principle of common fate. This strategy ensures that observations with similar trajectories are positioned in close proximity, creating cohesive visual bands. These bands facilitate the perception of distinct groups as they move together through high-dimensional space, reducing visual noise and highlighting underlying patterns.

Interaction with axis ordering. The effectiveness of hierarchical sorting is not independent of axis order. VanderPlas et al. (2023) note that the order of factor levels is an important determinant of residual line crossings: some crossings are attributable to the data structure itself and cannot be eliminated by within-category sorting, while others arise from an unfavorable factor ordering and can be removed by reordering levels. The `ggpcp` package allows users to reorder factor levels via `mutate` before variable selection, and to reverse axes to reduce negatively correlated crossings. Our tie-breaking extension inherits this dependency: for numerical axes, the minimum number of necessary crossings is a function of both the scaled data values and the sequence in which axes are processed. Users should therefore consider axis ordering as a precondition for optimal tie-breaking, particularly when integer-valued variables with many ties are adjacent to continuous variables with high variance.

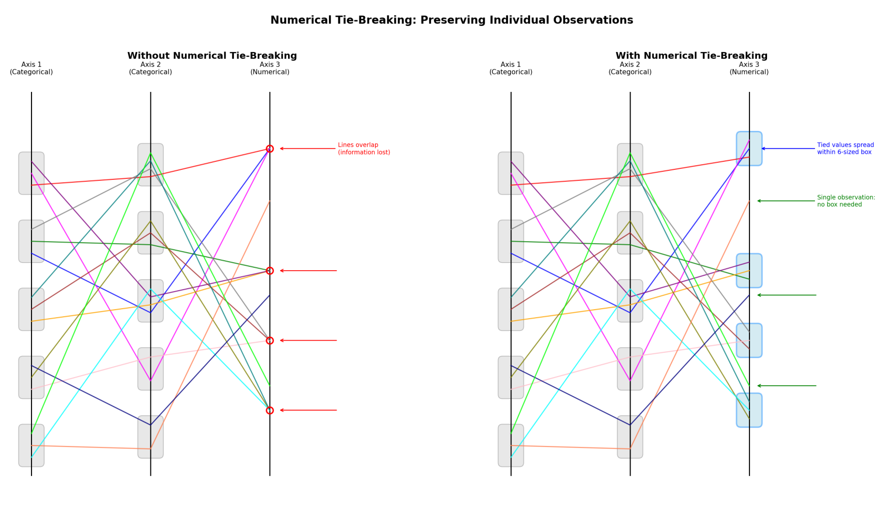

Figure 12: Comparison of parallel coordinate plots without (left) and with (right) numerical tie-breaking using `from-right` hierarchical sorting. When observations share the same numerical value, their lines overlap completely (left), hiding how many cases exist at that value. Spreading tied values within resolution-based boxes (blue) and sorting by the right neighbor axis (right) makes each observation visible, eliminates crossings within the tie band, and signals that positions within the box are artifacts of tie-breaking rather than meaningful data variation.

2.3 Gestalt Principles and Line Continuity

The principle of good continuation states that the human visual system preferentially perceives smooth, continuous contours over interpretations requiring abrupt changes in direction. This principle directly impacts the effectiveness of parallel coordinate plots for tracing individual observations.

Equispaced lines maintain continuity by representing each observation as a continuous polyline extending from the leftmost to the rightmost axis. When observations share identical numerical values (ties), the equispacing mechanism distributes lines within each category so that they remain visually distinct without introducing discontinuities.

Constant-width boxes represent observations as aggregated area segments rather than individual lines. This encoding communicates aggregate quantities effectively: the area of each segment is proportional to the number of observations sharing that combination of values across adjacent axes, making the joint frequencies of category pairs immediately visible. In contrast, equispaced lines preserve individual observations as distinct visual elements. When lines are continuous and uninterrupted, viewers can leverage preattentive processes such as the Gestalt principle of good continuation to trace a single case across all axes (Healey and Enns 2012). The two approaches thus serve distinct purposes: lines support individual-level tracing through low-level perceptual grouping, while boxes support aggregate-level frequency comparison. The conditions under which each is optimal are taken up in the Conclusion.

2.4 Visual Cue Design

The introduction committed to developing "a consistent visual representation to provide visual cues which indicate the presence of a local distortion due to tied values." This section specifies the design rationale for those cues, building on the perceptual principles outlined in the preceding section.

When tied values on a numerical axis are spread within a resolution-based interval, the positions of observations within that interval are artifacts of tie-breaking rather than reflections of meaningful data variation. A viewer who is unaware of this may misread the spread as genuine within-group variance. The visual cue must therefore accomplish two things simultaneously: it must make the presence of a tie region legible without disrupting the tracing of individual lines through the region.

The design adopted here uses a light gray bounding box, anchored at the lower and upper bounds of the resolution interval $[L(v), U(v)]$, as illustrated in Figure 14 and Figure 15. This choice follows directly from the categorical tie-breaking convention already implemented in `ggpcp`, where gray boxes group observations within each categorical level. Extending the same visual element to numerical tie regions creates a unified grammar: a gray box in any axis region signals "these positions are spread for legibility, not because the underlying values differ." This consistency reduces the cognitive overhead required to interpret mixed categorical-numerical plots.

Three alternative cue designs were considered and set aside. A color change on tied lines would require an additional channel already commonly used for group membership. A texture fill would introduce ambiguity on screen at small box heights. A dashed axis segment at the tied value would mark the location but not the extent of the spread, making it difficult to recover the original value. The gray box avoids all three problems: it marks both the location (the center of the box corresponds to the tied value) and the extent (the box height reflects δ_j, the global axis resolution), and it does not consume an encoding channel needed elsewhere.

The `get_tie_box()` function implements this detection and rendering pipeline in three steps: identifying values that appear more than once on a given axis, grouping contiguous spreads at numerical precision, and generating bounding box coordinates for `geom_rect` rendering. An optional label argument displays the original tied value at the center of each box, which is particularly informative for integer-valued variables where many values may be tied at round numbers.

3 Mathematical Framework

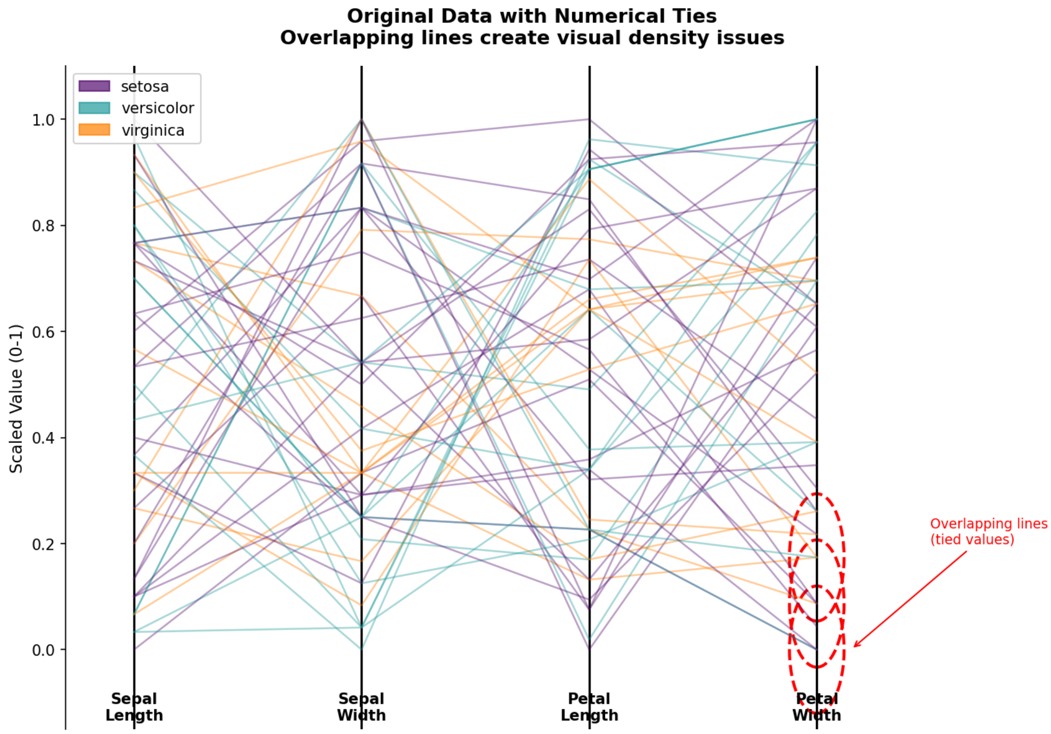

Figure 13: The tie problem in parallel coordinate plots. When observations share values, lines overlap and create visual collisions that mask the underlying data distribution.

3.1 Notation

Let $\mathcal{D}$ be a dataset with n observations and p variables displayed on parallel axes $X_1, X_2, \ldots, X_p$.

Table 1: Core notation

Symbol	Definition
x_{ij}	Raw value of observation i on variable j
y_{ij}	Scaled value (normalized to $[0, 1]$)
$\tilde{y}_{ij}$	Adjusted value after tie-breaking
$T_j(v)$	Set of observations tied at value v on axis j
k	Number of tied observations: $k = \lvert T_j(v) \rvert$
δ_j	Resolution of axis j: minimum distance between distinct values

3.2 Scaling

Variables are normalized to $[0,1]$ using min-max scaling:

$$y_{ij} = \frac{x_{ij} - \min(X_j)}{\max(X_j) - \min(X_j)} \tag{1}$$

3.3 The Tie-Breaking Algorithm

3.3.1 Step 1: Compute Axis Resolution

To preserve density information and maintain consistent spacing, we first compute the global resolution of axis j—the smallest difference between any two distinct values:

$$\delta_j = \min\{|y_1 - y_2| : y_1 \neq y_2,\ y_1, y_2 \in Y_j\} \tag{2}$$

where Y_j is the set of unique values on axis j.

Using a consistent jitter range across all values ensures that the visual representation preserves both the density of observations at each value and the gaps between distinct values. An approach that computed jitter ranges individually for each tied value would risk collapsing visually meaningful gaps, destroying distance information in the display.

3.3.2 Step 2: Identify Ties

A value v on axis j is a tie if multiple observations share it:

$$T_j(v) = \{i : y_{ij} = v\}, \quad |T_j(v)| > 1 \tag{3}$$

3.3.3 Step 3: Compute Available Space

Each value v is allocated a vertical interval on the axis bounded by the half-resolution distance to its neighbors:

$$L(v) = v - \frac{\delta_j}{2}, \quad U(v) = v + \frac{\delta_j}{2} \tag{4}$$

If v is at the axis boundary, apply a buffer β (default 0.05):

- $L(v) = \max(0, v - \delta_j/2)$ if $v - \delta_j/2 < 0$
- $U(v) = \min(1, v + \delta_j/2)$ if $v + \delta_j/2 > 1$

The total allocation per value is:

$$S(v) = U(v) - L(v) = \delta_j \tag{5}$$

This $S(v)$ is the numerical analog of the categorical S_i: it is the full per-unit allocation on the axis, including the margins reserved for inter-unit separation.

Separation constraint. Allocating the entire interval $[L(v), U(v)]$ to spread observations would, for evenly-spaced distinct values, produce regions that share endpoints with their neighbors and render as a single continuous column on the axis. This collapses the visual separation between distinct values and defeats the Gestalt grouping cue (see Figure 8). We therefore introduce a separation parameter $\sigma \in (0, 1]$ (default $\sigma = 0.9$, provisional; see Section 3.3.5) that reserves margins of equal width above and below the dot-placement region:

$$S^+(v) = S^-(v) = \frac{(1-\sigma) \cdot \delta_j}{2} \tag{6}$$

These margins are the numerical analog of the categorical S_i^+ and S_i^-: where the categorical margins are user-controlled via the `space` argument, the numerical margins are derived from σ. The dot-placement region — the actual vertical extent within which tied observations are spread — is the residual:

$$S(v) - S^+(v) - S^-(v) = \sigma \cdot \delta_j \tag{7}$$

with bounds $[L(v) + S^-(v), U(v) - S^+(v)]$. The unencoded margins on either side preserve the perceptual gap between adjacent value bands.

3.3.4 Step 4: Compute Optimal Spacing

The inter-observation spacing is the dot-placement region (Equation Equation 7) divided into $k-1$ intervals:

$$\boxed{\Delta(v) = \frac{S(v) - S^+(v) - S^-(v)}{k-1} = \frac{\sigma \cdot \delta_j}{k-1}} \tag{8}$$

This is the same formula used for categorical levels:

$$d_i = \frac{S_i - S_i^+ - S_i^-}{n_i - 1}$$

The two formulas are structurally identical; the innovation is in what determines S and $S^{\pm}$. For categorical levels, S_i is set by frequency proportion under a user-chosen total budget. For numerical values, $S(v) = \delta_j$ is set by the global axis resolution and is the same for every value v, while $S^{\pm}(v)$ is derived from σ. The unified spacing principle — *spacing equals usable space divided by* $n - 1$ — applies in both cases.

3.3.5 How Much Spread Is Too Much?

The spacing formula $\Delta(v) = \sigma \cdot \delta_j/(k-1)$ treats the dot-placement region — the inner band of height $\sigma \cdot \delta_j$ — as given. The choice of σ determines how far an individual observation may be displaced from the true value v before the displacement is read as genuine variation rather than as an artifact of tie-breaking. This is the central empirical question raised by the proposal and is not resolved by the algorithm as currently specified.

Two competing pressures define the admissible range of σ. From below, σ must be large enough that adjacent observations within the band are perceptually distinguishable; as $\sigma \to 0$ the band collapses to a point and the spread no longer accomplishes its purpose. From above, σ must be small enough that the band does not encroach on the gap between adjacent values. The hard upper limit $\sigma \leq 1$ enforces the latter mathematically; in practice, the perceptually defensible upper limit is strictly smaller, because the reader must *see* the gap, not merely that it exists in the coordinate system.

Where in the interval $(0, 1)$ does σ belong? The value $\sigma = 0.9$ adopted elsewhere in this proposal is a working default rather than a derived optimum. We propose to treat σ as an empirical parameter and to determine its admissible range using the same machinery that has been used to bound related perceptual distortions in parallel-axis displays: the sine illusion work of VanderPlas and Hofmann (2015), the line-width illusion analysis of Hofmann and Vendettuoli (2013), and the broader Müller-Lyer literature on which both papers draw. The sine illusion in particular establishes that vertical displacement is not perceived neutrally; line segments of equal length appear unequal when arrayed along a curve, with the distortion quantifiable as a function of local slope. The same perceptual mechanism that produces the sine illusion bounds how much within-band displacement a viewer can sustain before reading the displacement as real variance.

We therefore frame the tolerance question as a research question in its own right: given a target task, frequency estimation, individual tracing, or outlier detection, what is the largest σ for which task performance is statistically indistinguishable from performance at $\sigma = 0$? This is testable within the perceptual validation phase outlined in Section 4.2, and the σ that survives across tasks is the σ we recommend as a default. The methodology and sample-size calculation for the parameter-sweep study appear in Section 4.2 as Study 1b.

3.3.6 Within-Band Rank and Deviation as Outlier Signals

A consequence of the hierarchical sorting in Step 5 is that an observation's position within its tie band is not noise. Under `from-right` processing, the observation placed at the lowest position in the band is the one with the lowest adjusted value on the right-neighbor axis; the observation placed at the highest position is the one with the highest. Within-band rank therefore carries information about the right-neighbor axis, and the deviation of any observation from the band centroid is a direct visual encoding of how far that observation lies from the median of its tie group on the next axis.

This makes the band a low-cost outlier-detection device. An observation that sits at the extreme of a tie band, and continues to extreme positions on neighboring axes, is a candidate outlier whose status can be confirmed by tracing the polyline. An observation that sits near the centroid of its band and connects to centroid positions elsewhere is, by construction, near the joint mode of the data. We do not propose a formal outlier statistic in this proposal, but we note that the rank-and-deviation structure produced by the algorithm is a natural object for one, and we expect this to be a productive direction for follow-on work.

3.3.7 Step 5: Order Observations Hierarchically

To minimize line crossings, tied observations are sorted by their positions on the adjacent axis before spread positions are assigned. For axis j, let A_j be the adjacent axis, the left neighbor under `from-left` processing, or the right neighbor under `from-right` processing.

For each observation $i \in T_j(v)$, the sorting key is:

$$\kappa_i = \tilde{y}_{i,A_j} \tag{9}$$

Observations are ordered so that $\kappa_{i_1} \leq \kappa_{i_2} \leq \cdots \leq \kappa_{i_k}$.

This ordering is the step that resolves the visual "knots" illustrated in Figure 8. Spreading alone distributes tied points across the interval $[L(v), U(v)]$, but without a consistent rank-matching to the neighbor axis, lines departing the band will cross inside it. Sorting by κ_i before assigning positions in Equation 10 creates a monotone mapping: the observation placed at rank m in the tie band is exactly the observation ranked m-th on the neighbor axis. Under `from-right` processing, this means the lowest position in the band connects to the lowest position on the right axis, eliminating within-band crossings entirely. Any crossings that remain after this step are necessary crossings induced by the data structure itself.

Key insight: This requires *sequential* axis processing, the adjusted positions from previously processed axes inform the ordering on the current axis.

3.3.8 Step 6: Assign Positions

Given ordered observations $(i_1, i_2, \dots, i_k)$, positions are assigned within the dot-placement region, the interior of the allocation $[L(v), U(v)]$ after the margins $S^-(v)$ and $S^+(v)$ are reserved:

$$\tilde{y}_{i_m,j} = L(v) + S^-(v) + (m-1) \cdot \Delta(v), \quad m = 1, 2, \dots, k \tag{10}$$

3.4 Unified Framework

The algorithm treats numerical values and categorical levels identically at the formula level: the same spacing equation governs both, and the same five quantities (S, S^+, S^-, n, and the inter-observation spacing) are present in both cases. What differs is how each quantity is determined.

Table 2: Unified spacing framework: identical formula structure, type-specific parameter semantics.

	Categorical	Numerical
Total allocation	S_i (set by frequency proportion under user budget)	$S(v) = \delta_j$ (set by axis resolution; constant across values)
Margin above	S_i^+ (user-controlled via `space`)	$S^+(v) = (1-\sigma) \cdot \delta_j/2$ (derived from σ)
Margin below	S_i^- (user-controlled via `space`)	$S^-(v) = (1-\sigma) \cdot \delta_j/2$ (derived from σ)
Count	n_i (level frequency)	$n(v) = k$ (tie multiplicity)
Spacing	$d_i = (S_i - S_i^+ - S_i^-)/(n_i - 1)$	$\Delta(v) = (S(v) - S^+(v) - S^-(v))/(n(v) - 1)$

The shared formula is **spacing = (allocation − margins) / (n − 1)**. The innovation in extending the categorical procedure to numerical variables lies entirely in the right-hand column:

anchoring the per-value allocation to the axis resolution and deriving the margins from a single perceptual parameter σ, rather than from frequencies and user input.

3.5 Visual Results

3.5.1 Uniform Spacing Applied

After applying the tie-breaking algorithm, observations that previously overlapped are now visually distinguishable:

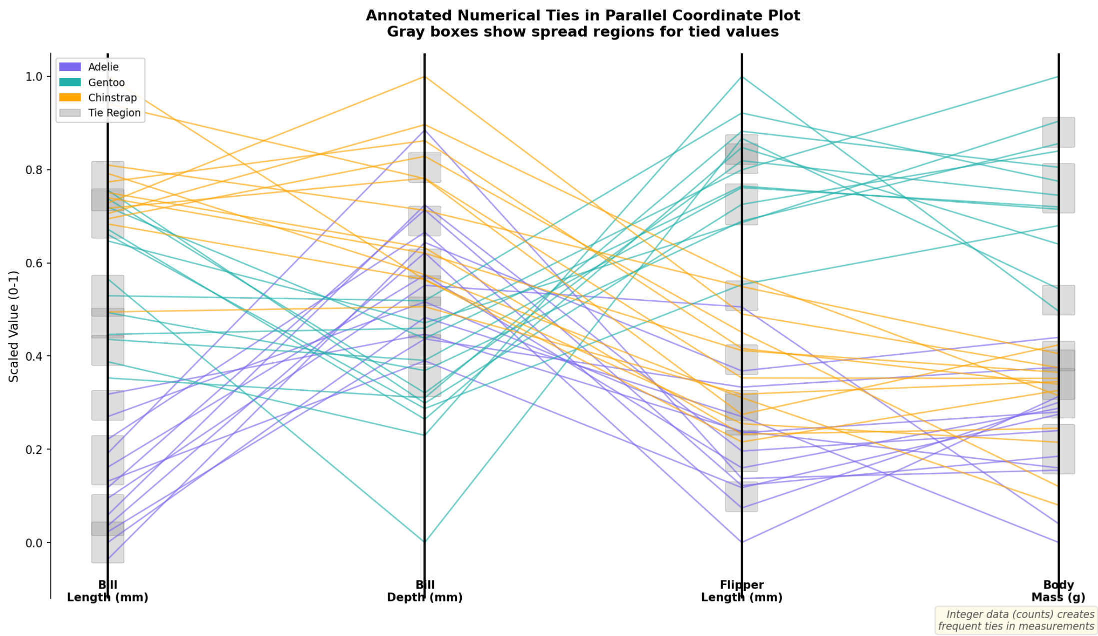

Figure 14: Parallel coordinate plot after applying uniform spacing. Gray boxes indicate regions where tied values have been spread, making individual observations traceable.

3.5.1.1 Annotated Tie Regions

Tie regions can be annotated with boxes showing the original value and the spread region:

3.5.2 Tie Box Detection

The `get_tie_box()` function identifies and bounds tie regions for visualization:

3.6 Algorithm Summary

Algorithm: Unified Tie-Breaking

Input: Data D, variables $X_1 ... X_p$, method $\in$ {from-left, from-right, from-both}

Output: Adjusted positions $_{ij}$

1. Scale all variables to $[0, 1]$
2. For each axis j, compute resolution: $\delta_i = min|y_1 - y_2|$ for distinct $y_1, y_2 \in Y_j$
3. Set processing order based on method:

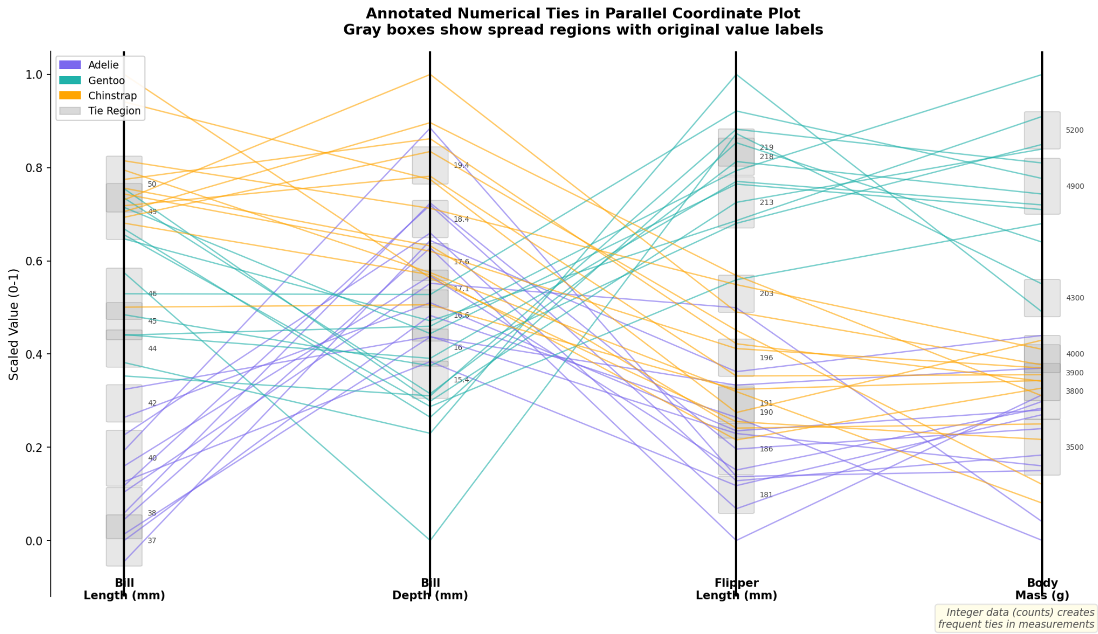

Figure 15: Annotated tie regions with labels. Gray boxes highlight where ties occurred, with labels showing the original tied value. This is particularly useful for integer data where ties are frequent.

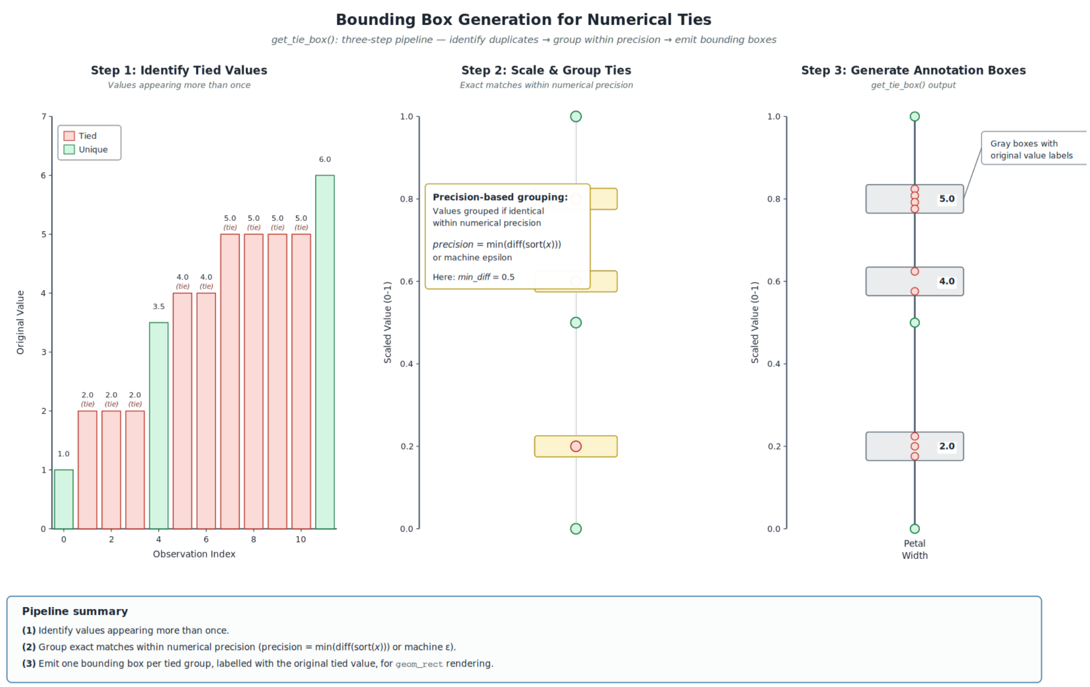

Figure 16: The tie box detection process. (1) Identify duplicated values, (2) scale and group contiguous spreads, (3) generate bounding box coordinates for annotation.

- from-left: process axes $1, 2, \ldots, p$; reference left neighbor
- from-right: process axes $p, p-1, \ldots, 1$; reference right neighbor
- from-both: process outward from center axis; reference previously processed neighbor

4. For each axis j in processing order:
   a. Identify tied values: $V = \{v : |T_j(v)| > 1\}$
   b. For each $v \in V$:
      i. Set total allocation: $S(v) = \delta_j$
      ii. Compute outer bounds: $L(v) = v - \delta_j/2$, $U(v) = v + \delta_j/2$
      iii. Set margins: $S^+(v) = S^-(v) = (1-\sigma) \cdot \delta_j/2$
      iv. Order $T_j(v)$ by $\kappa_i = \tilde{y}_{i,A_j}$ (positions on adjacent axis; right neighbor under `from-right`, left neighbor under `from-left`)
      v. Compute spacing: $\Delta(v) = (S(v) - S^+(v) - S^-(v))/(k-1)$
      vi. Assign positions: $\tilde{y}_{i_m,j} = L(v) + S^-(v) + (m-1) \cdot \Delta(v)$ for $m = 1, \ldots, k$
5. Return adjusted data

3.7 Parameters

Table 3: Algorithm parameters

Parameter	Symbol	Default	Description
Band separation	σ	0.90 (provisional)	Fraction of δ_j used as band width; $(1-\sigma) \cdot \delta_j$ is preserved as gap between adjacent bands. The default is a working value to be revised based on the perceptual study described in Section 3.3.5 and Study 1b in Section 4.2. The admissible range is bounded below by the within-band distinguishability requirement and bounded above by the Gestalt-of-proximity requirement that the inter-band gap remain visible; the empirical study determines where in this range σ belongs.
Boundary buffer	β	0.05	Fallback when no neighbor exists
Minimum spacing	δ	0.01	Ensures minimum separation
Tie tolerance	ε	10^{-10}	For floating-point comparison

4 Implementation and Evaluation Plan

4.1 Phase 1: Algorithm Development (Weeks 1-4)

Deliverable: Functional R package extension with comprehensive documentation Tasks:

1. Extend `pcp_arrange()` to detect and handle numerical ties
2. Implement `tie_spacing_numerical()` function with perceptually-motivated parameters anchored to the global axis resolution δ_j and the band-separation parameter σ
3. Handle edge cases while preserving perceptual properties:

- Single unique value (centered position)
- Extreme skew (adaptive buffer sizing)
- Missing values (explicit separation region)

4. Integration testing with existing ggpcp workflow

Code Structure (Simplified):

```
# Resolution-anchored numerical tie-spreading.
# `values` is a numeric vector of *scaled* values in [0, 1] (one per observation).
# Returns one adjusted position per observation, in the input order.
tie_spacing_numerical <- function(values, sigma = 0.9, beta = 0.05) {
  # 1. Identify unique values and the global axis resolution
  unique_vals <- sort(unique(values))
  m           <- length(unique_vals)
  if (m < 2) {
    # Single unique value: centre at the value itself
    return(rep(unique_vals[1], length(values)))
  }
  delta_j <- min(diff(unique_vals))   # smallest gap between distinct values

  # 2. Per-value allocation S(v) = delta_j; inner spread region = sigma * delta_j
  inner_half <- (sigma * delta_j) / 2

  # 3. Position observations within each tied group
  positions <- numeric(length(values))
  for (v in unique_vals) {
    idx <- which(values == v)
    n_v <- length(idx)

    # Boundary handling: clip the inner region if v is near the [0, 1] edges
    L_inner <- max(0, v - inner_half)
    U_inner <- min(1, v + inner_half)
    # If clipping eliminated the inner region (rare), fall back to buffer beta
    if (U_inner - L_inner <= 0) {
      L_inner <- max(0, v - beta)
      U_inner <- min(1, v + beta)
    }

    if (n_v == 1) {
```

```
      positions[idx] <- v
    } else {
      # Spread n_v observations from L_inner to U_inner, endpoints inclusive
      positions[idx] <- seq(L_inner, U_inner, length.out = n_v)
    }
  }

  return(positions)
}
```

The full implementation in `pcp_arrange()` adds the hierarchical-sorting step (Step 5 of the algorithm) before `seq()` is called, so observations are placed in the order given by their adjusted positions on the neighboring axis. The simplified stub above isolates the spacing logic; the sorting step is documented separately in Section 2.2.

4.2 Phase 2: Perceptual Validation Studies (Weeks 5–7)

Success Criteria. Before detailing the study design, we specify the criteria that would constitute evidence of success for RQ2 and RQ3. For RQ2, we consider the algorithm successful if GPCP with tie-breaking produces an absolute percentage error on magnitude estimation tasks that is significantly lower (two-tailed $\alpha = 0.05$) than the hammock plot baseline, with $d \geq 0.4$. For ordinal and ratio tasks, we require accuracy rates at least 10 percentage points higher than baseline. For RQ3, we require that tracing accuracy in GPCP with tie-breaking exceeds that of the standard PCP without tie-breaking by at least 15 percentage points. Outcomes that do not meet these thresholds will be interpreted as evidence that the algorithm achieves legibility parity but does not strictly surpass the comparison conditions.

Study 1b: Determining the admissible σ.

A precondition for interpreting Studies 1 and 2 is that the band-width parameter σ is set at a value that does not itself bias frequency or tracing judgments. We add a parameter-sweep study, $n = 60$ (between-subjects, 12 participants per σ level), to determine the largest σ for which task performance is statistically indistinguishable from $\sigma = 0$.

Procedure. Participants complete the magnitude-estimation and ordinal-comparison battery used in Study 1, on stimuli generated at $\sigma \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ with axis resolution and tie-frequency structure held fixed across conditions.

Analysis. Per-task absolute error is regressed on σ; the largest σ at which the 95% confidence interval on error overlaps the $\sigma = 0$ estimate is taken as the perceptually admissible upper bound. We pre-register this as a non-inferiority criterion with margin equal to half the smallest effect detectable in the main Study 1 power analysis.

Connection to sine-illusion bounds. Where our empirically determined σ falls relative to the displacement bounds estimated by VanderPlas and Hofmann (2015) is itself informative. Agreement strengthens the perceptual argument for the spread approach; disagreement points to a context-specific factor — axis density, line continuity, or the categorical-numerical interaction — that warrants further investigation beyond the scope of this proposal.

Study 1: Frequency Perception

- Duration: Weeks 5–7
- Participants: n = 128 (64 per condition)
- Design: Between-subjects comparison of GPCP vs. hammock plot representations

Sample Size Justification:

A power analysis was conducted using G*Power 3.1 (Faul et al. 2007) to determine the minimum sample size required to detect a meaningful difference between visualization conditions. Based on prior graphical perception studies—including Heer and Bostock's (2010) crowdsourced replication of Cleveland and McGill (1984), which used approximately 50 participants per condition, and Hofmann and Vendettuoli's (2013) common angle plot study with 46 participants, we anticipated a medium effect size (Cohen's $d = 0.5$). For a two-tailed independent samples t-test with $\alpha = 0.05$ and $power = 0.80$, the required sample size is $n = 64$ per group (total $N = 128$). This sample size also provides 80% power to detect effects as small as $d = 0.50$, which represents a practically meaningful difference in task accuracy between visualization types (Brysbaert 2019). To account for potential exclusions due to failed attention checks or incomplete responses (estimated at 10–15% based on crowdsourcing norms), we will recruit $n = 75$ per condition.

Procedure:

1. Magnitude estimation (12 trials): "What percentage have value X?"
2. Ordinal comparison (12 trials): "Which value is more frequent?"
3. Ratio judgment (12 trials): "A is how many times B?"
4. Confidence ratings (7-point scale) for each response

Analysis:

- Absolute percentage error for magnitude estimation
- Accuracy rates for ordinal and ratio tasks
- Bias analysis: systematic over/under-estimation
- Confidence calibration: accuracy vs. subjective confidence
- Effect sizes reported as Cohen's d with 95% confidence intervals

4.3 Phase 3: Comparative Benchmarking (Weeks 6–10)

Visual Clutter and Information Density. Visual clutter constitutes a fundamental limitation on visualization effectiveness, occurring when too many visual elements compete for attention and overwhelm the viewer's capacity to extract meaningful patterns. The two approaches manage clutter through fundamentally different mechanisms. For equispaced lines, clutter increases with the number of observations, but hierarchical sorting minimizes line crossings and thereby reduces visual complexity. The approach exhibits graceful degradation: as density increases, individual lines transition perceptually into ribbons and eventually into filled areas, yet individual observations remain theoretically accessible for highlighting or interactive selection. For constant-width boxes, clutter depends primarily on the number of unique value combinations rather than on raw observation counts. Aggregation inherently reduces clutter, though at the cost of individual-level accessibility.

Dennig et al. (2021) formalize a suite of clutter metrics for Parallel Sets visualizations, including ribbon overlap, crossing angle, orthogonality, and ribbon width variance, and demonstrate that these metrics are mutually uncorrelated, measuring distinct properties of the display. These metrics can be adapted directly to compare the two approaches quantitatively here, and to guide the selection of axis and category orderings that minimize visual complexity for each dataset.

Computational Metrics:

1. Rendering performance (time, memory, scalability n = 2 to 100)
2. Visual quality metrics (Dennig et al. 2021): line crossing count, ribbon overlap, visual clutter index
3. Perceptual quality estimates: modeled eye movements, predicted visual search time

Case Studies:

- Palmer Penguins: Mixed categorical-numerical data with known correlation structure
- Iris Dataset: Multiple numerical variables with natural ties
- Asthma Data (Schonlau and Yang 2024): Direct comparison with published hammock plot

4.4 Phase 4: Integration and Dissemination (Weeks 8–12)

Deliverables:

1. R Package Update:

- CRAN submission with full documentation
- Vignette: "Handling Numerical Ties in Parallel Coordinate Plots"
- Unit tests achieving $> 95\%$ coverage

2. Academic Paper:

- Target: IEEE Transactions on Visualization and Computer Graphics
- Submission deadline: August 2026 (IEEE VIS)

3. Supplementary Materials:

- Open Science Framework repository with pre-registration
- All experimental stimuli and data
- Reproducibility package with power analysis scripts

5 Limitations and Future Directions

5.1 Study Limitations

- **Sample Characteristics:** University student population may not generalize to domain experts
- **Experimental Constraints:** Laboratory tasks may lack ecological validity
- **Scope:** Limited to static visualizations; interactive features not evaluated

5.2 Future Extensions

- Longitudinal study with domain experts in real analytical workflows
- Investigation of interaction effects between individual differences and visualization method
- Hybrid approaches: smooth interpolation between lines and boxes based on dataset properties
- Integration with animation and interactive highlighting techniques

6 Conclusion

Leveraging tie-breaking and sorting for numerical variables addresses a known issue with GPCPs while maximizing the numerical scale information available to the reader. The proposed equispaced line method extends ggpcp's established categorical tie-breaking algorithm to numerical variables, creating a unified framework for handling ties across variable types. Implementing numerical tie-breaking preserves the Gestalt continuity of a single observation across parallel axes and allows the viewer to more easily recreate the joint density across multiple axes, relative to the standard numerical axis approach.

It is worth distinguishing between two uses of box-like visual elements in parallel coordinate displays. Hammock plots use parallelograms to represent aggregated bivariate relationships, where the area of each segment encodes the joint frequency of category pairs across adjacent axes. This approach scales well with large datasets but does not permit tracing of individual observations. GPCPs take a different approach: when multiple observations share the same value, their vertical positions are spread within a frame defined by the global resolution of the axis—the smallest difference between any two distinct values. This frame signals to the reader that positions within it reflect tie-breaking rather than meaningful data variation, while individual lines remain continuous and traceable.

Rather than viewing these as competing representations, they can be understood as complementary. The GPCP approach is optimal when the analytical task involves following specific cases or detecting outliers, whereas the hammock approach is optimal when the task involves comparing category frequencies or understanding distributional patterns without regard to individual observations. With many observations, equispaced lines may coalesce into ribbon-like bands; however, the principle of good continuation still applies, allowing viewers to perceive flow patterns even when individual lines are indistinguishable.

This research contributes to theory by demonstrating how Gestalt principles and models of preattentive processing generate testable hypotheses about visualization effectiveness. It contributes to practice by providing open-source software and evidence-based design guidelines. And it contributes to methodology by illustrating theory-driven, multi-method evaluation—computational benchmarking to verify visual quality before perceptual testing, followed by appropriately powered user studies grounded in the graphical perception literature.

Effective visualization must be rooted in how people actually perceive: how they use principles of proximity, similarity, continuity, and common fate to organize and interpret visual features. This study takes that principle seriously by asking not merely "which visualization looks better?" but "which visualization aligns more effectively with perceptual organization principles for specific analytical tasks?" The answer, as with most questions in visualization, will depend on context. But it will now rest on empirical evidence and perceptual theory rather than assumption.

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